

R Package lessR

Way More Analysis with Much Less Work

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Overview

- ▶ The lessR package simplifies analysis and visualization.
- ▶ Focuses on minimal code with comprehensive output.
- ▶ A minimal number of functions to:
 - ▶ Focus on parameters for output control instead of more function calls
 - ▶ Serve as a common interface for otherwise disparate functions from Base R and user contributed packages
 - ▶ Bottom-up vs. top-down approach, where the default is to provide a reasonably complete solution instead of building step-by-step
 - ▶ A complete analysis implies visualizations plus statistics

Available Data File Formats in lessR

- ▶ One syntax to read data: `Read("location")`, `Read("")` browse
- ▶ One syntax to write data:
`Write(df_name, "location", format=fmt)`

Table: lessR `Read()` and `Write()` data file formats.

Extension	Format	Package	Source
.csv	text, custom separator	R utils	<code>read.csv()</code>
.tsv	text, tab separated values	R utils	<code>read.delim()</code>
.prn	text, space separated values	R utils	<code>read.table()</code>
.txt	text, comma or tab separated	R utils	<code>read.table()</code>
.xls, .xlsx	Excel	openxlsx	<code>read.xlsx()</code>
.ods	ODS	readODS	<code>read_ODS()</code>
.feather	feather	arrow	<code>read_feather()</code>
.parquet	parquet	arrow	<code>read_parquet()</code>
.rda	R	R base	<code>load()</code>
.sav, .zsav	SPSS	haven	<code>read_spss()</code>
.dta	Stata	haven	<code>read_stata()</code>
.sas7bdat	SAS	haven	<code>read_sas()</code>

Read Data into a Data Frame

► `d <- Read("Employee")`

>>> Suggestions

Recommended binary format for data files: feather

Create with `Write(d, "your_file", format="feather")`

More details about your data, Enter: `details()` for `d`, or `details(name)`

Data Types

character: Non-numeric data values

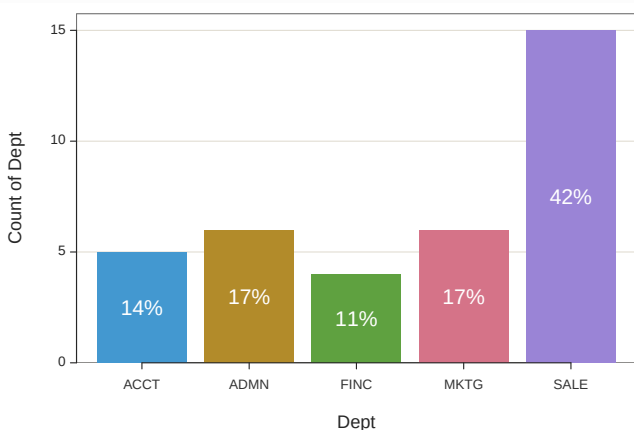
integer: Numeric data values, integers only

double: Numeric data values with decimal digits

	Variable Name	Type	Values	Missing Values	Unique Values	First and last values									
1	Years	integer	36	1	16	7	NA	7	...	1	10				
2	Gender	character	37	0	2	M	M	W	...	W	M				
3	Dept	character	36	1	5	ADMN	SALE	FINC	...	MKTG	FINC				
4	Salary	double	37	0	37	63788.26	104494.58	...	67562.3						
5	JobSat	character	35	2	3	med	low	high	...	high	low	hi			
6	Plan	integer	37	0	3	1	1	2	...	2	2	1			
7	Pre	integer	37	0	27	82	62	90	...	83	59	80			
8	Post	integer	37	0	22	92	74	86	...	90	71	87			

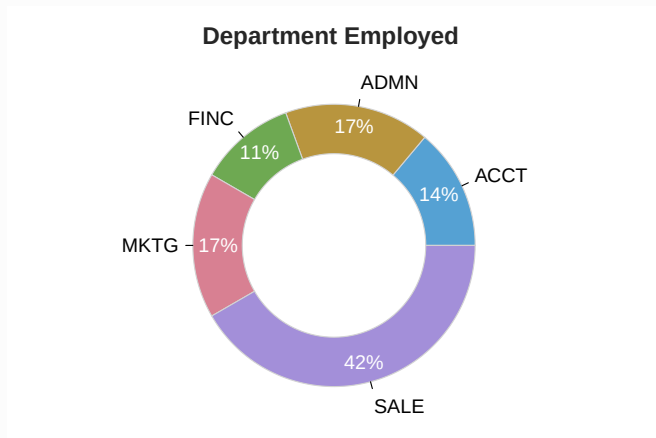
Default One Categorical Variable Bar Chart

► `BarChart(Dept)`



Default Pie Chart

► PieChart(Dept)



BarChart()/PieChart() Statistical Output

--- Dept ---

Missing Values: 1

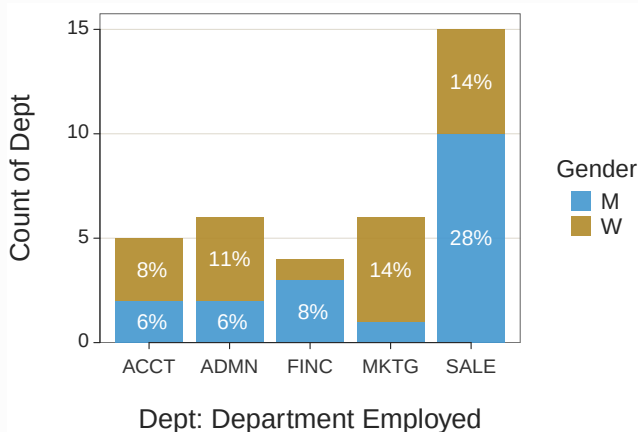
	ACCT	ADMN	FINC	MKTG	SALE	Total
Frequencies:	5	6	4	6	15	36
Proportions:	0.139	0.167	0.111	0.167	0.417	1.000

Chi-squared test of null hypothesis of equal probabilities

Chisq = 10.944, df = 4, p-value = 0.027

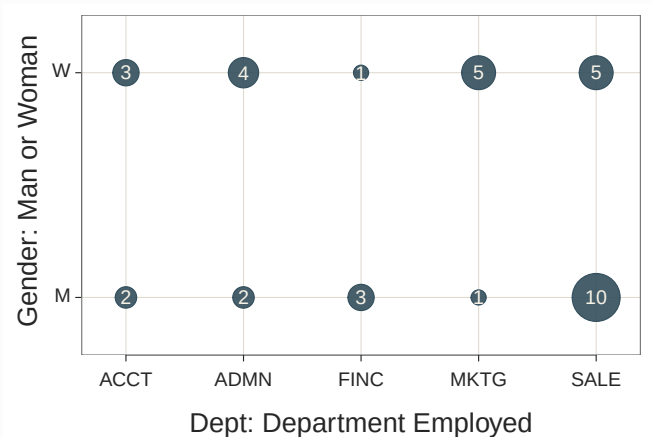
Default Two Categorical Variable Bar Chart

► `BarChart(Dept, by=Gender)`



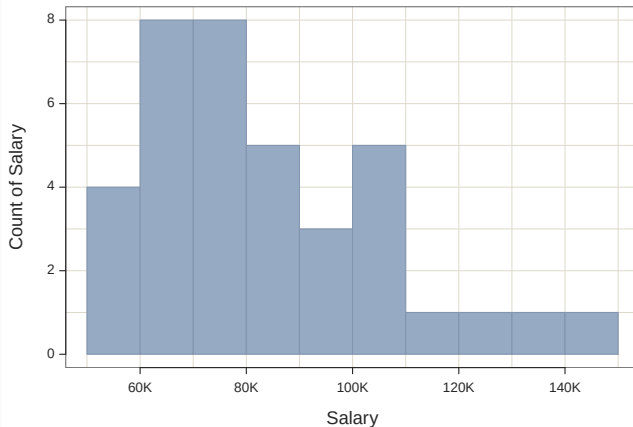
Default Two Categorical Variable Bubble Chart

- ▶ Scatterplot of two categorical variables
- ▶ `Plot`(Dept, Gender)



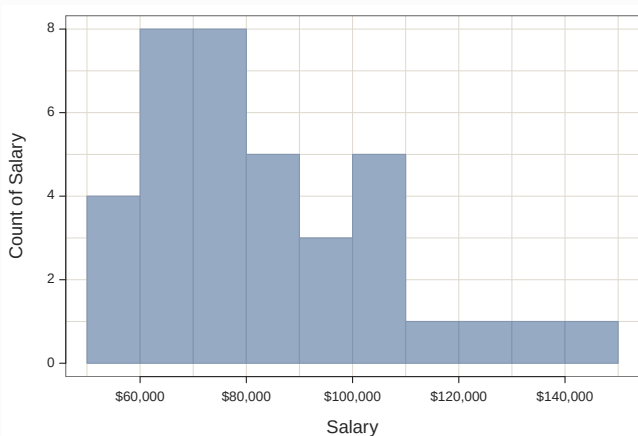
Default Histogram

► Histogram(Salary)



Histogram with Reformatted Axis Values

► `Histogram(Salary, axis_fmt="," , axis_x_pre="$")`



Histogram Statistical Output I

► — Salary —

n	miss	mean	sd	min	mdn	max
37	0	83795.56	21799.53	56124.97	79547.60	144419.23

► — Outliers — from the box plot: 1

Small	Large
-----	-----
	144419.2

Histogram Statistical Output II

Bin Width: 10000

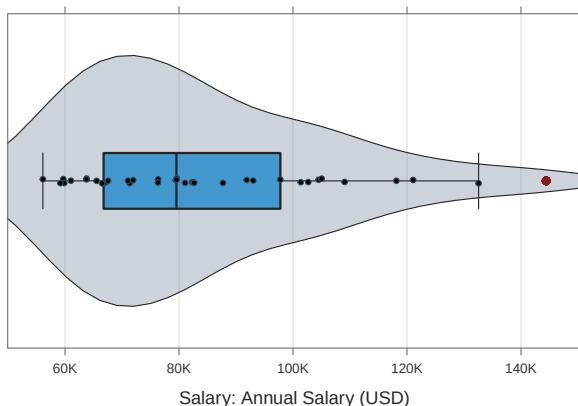
Number of Bins: 10

	Bin	Midpnt	Count	Prop	Cumul.c	Cumul.p

50000 >	60000	55000	4	0.11	4	0.11
60000 >	70000	65000	8	0.22	12	0.32
70000 >	80000	75000	8	0.22	20	0.54
80000 >	90000	85000	5	0.14	25	0.68
90000 >	100000	95000	3	0.08	28	0.76
100000 >	110000	105000	5	0.14	33	0.89
110000 >	120000	115000	1	0.03	34	0.92
120000 >	130000	125000	1	0.03	35	0.95
130000 >	140000	135000	1	0.03	36	0.97
140000 >	150000	145000	1	0.03	37	1.00

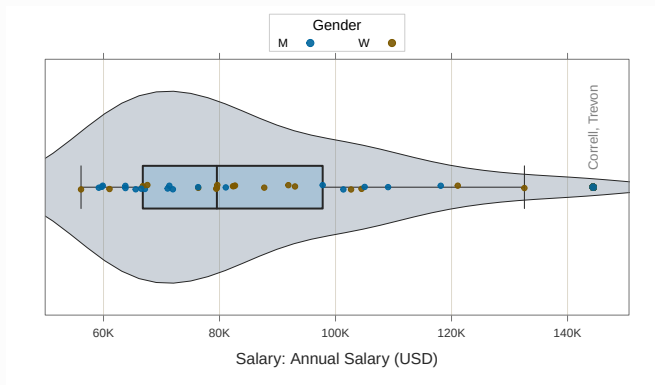
Say Goodbye to 19th Century Technology

- ▶ **VBS plot**: Integrated violin plot, box plot, scatter plot
- ▶ All parameters implicitly and automatically tuned:
 jitter, point size, bandwidth
- ▶ **Plot(Salary)**



Group Within-Panel: Parameter by

► Plot(Salary, by=Gender)



Group Between-Panels: Parameter facet1

► Plot(Salary, facet1=Gender)



VBS Console Output

[Trellis (facet) graphics from Deepayan Sarkar's lattice package]

>>> Suggestions

Plot(Salary, fences=TRUE, vbs_mean=TRUE) # Show inner fences Show mean

Plot(Salary, box_adj=TRUE) # Adjust boxplot whiskers for asymmetry

ttest(Salary ~ Gender) # Add the data parameter if not the d data frame

----- Pivot table for Salary

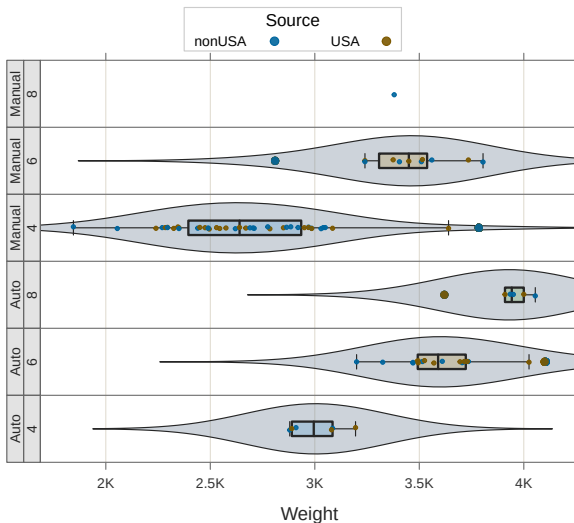
Gender	n	na	Mean	Median	SD	IQR	Min	Max
M	18	0	91147.46	89792.95	23128.44	27772.74	59188.96	144419.2
W	19	0	76830.60	71356.69	18438.46	15766.48	56124.97	132563.4

----- Parameter values (can be manually set)

size: 0.56	size of plotted points
out_size: 0.80	size of plotted outlier points
jitter_y: 0.54	random vertical movement of points
jitter_x:	random horizontal movement of points
bw: 9529.04	set bandwidth higher for smoother edges

VBS Four Variables

- `Plot(Weight, by=Source, facet1=Cylinders, facet2=TranType)`



VBS Four Variable Console Output

[Trellis (facet) graphics from Deepayan Sarkar's lattice package]

----- Pivot tables for Weight

Source	Cylinders	TranType	n	na	Mean	Median	SD	IQR	Min
USA	4	Auto	6	0	3006.667	2995.0	131.136	188.75	2880
nonUSA	6	Auto	5	0	3658.000	3730.0	329.367	210.00	3200
USA	6	Auto	15	0	3626.667	3570.0	210.998	217.50	3325
nonUSA	8	Auto	1	0	4000.000	4000.0	NA	0.00	4000
USA	8	Auto	5	0	3894.000	3935.0	162.842	40.00	3620
nonUSA	4	Manual	27	0	2745.741	2775.0	414.111	590.00	2055
USA	4	Manual	16	0	2538.438	2552.5	268.984	270.00	1845
nonUSA	6	Manual	6	0	3362.500	3457.5	279.728	131.25	2810
USA	6	Manual	5	0	3494.000	3450.0	267.287	495.00	3240
USA	8	Manual	1	0	3380.000	3380.0	NA	0.00	3380
nonUSA	NA	Manual	6	0	2634.167	2470.0	880.445	1172.50	1695

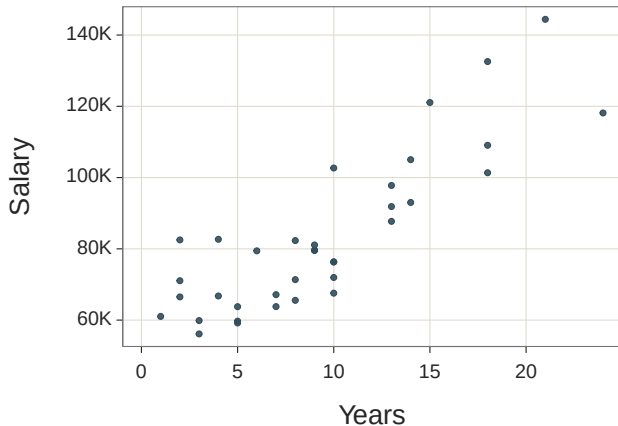
Source	n	na	Mean	Median	SD	IQR	Min	Max
nonUSA	45	0	2942.333	2950.0	593.753	930.00	1695	4100
USA	48	0	3195.312	3282.5	565.228	933.75	1845	4105

...

and for Cylinders and TranType

Default ScatterPlot

► `Plot(Years, Salary)`



ScatterPlot Statistical Output

```
>>> Pearson's product-moment correlation
```

```
Number of paired values with neither missing, n = 36
```

```
Sample Correlation of Years and Salary: r = 0.852
```

```
Hypothesis Test of 0 Correlation:
```

```
    t = 9.501,  df = 34,  p-value = 0.000
```

```
95% Confidence Interval for Correlation:
```

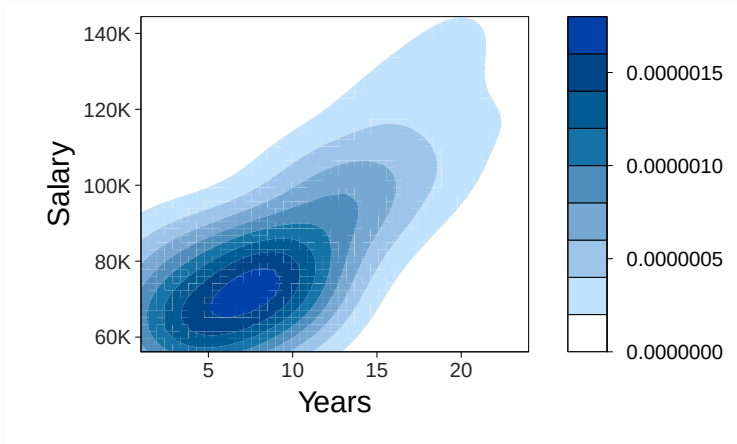
```
    0.727 to 0.923
```

Enhanced ScatterPlot

- ▶ Each of the following can be added individually or all at once
-
- ▶ 95% confidence ellipse
 - ▶ best-fit least squares line
 - ▶ outlier detection and labeling based on Mahalanobis distance
 - ▶ best-fit least squares line without the outliers
 - ▶ vertical line of mean of x and horizontal line of the mean of y

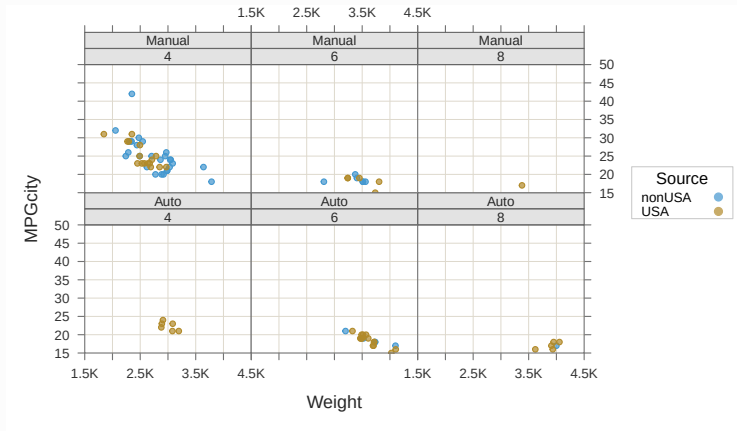
Contour Plot

- ▶ We do not need to assume multivariate normality to generalize to the bivariate densities, even for n of only 37
- ▶ `Plot(Years, Salary, type="contour")`



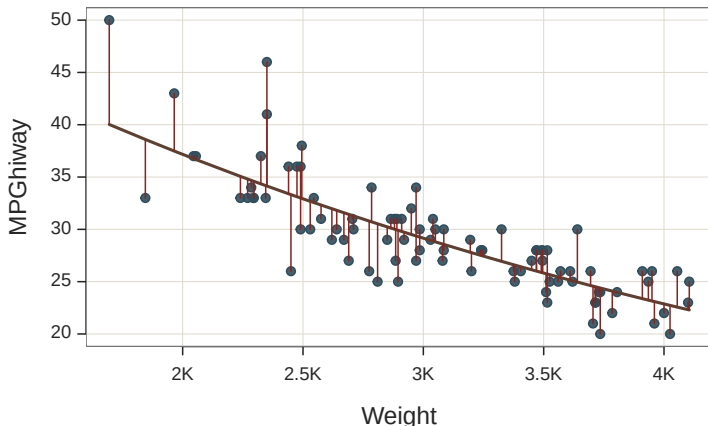
Scatterplots: Facets and Group-By Categorical Variables

- ▶ `d <- Read("Cars93")`
- ▶ `Plot(Weight, MPGcity, by=Source, facet1=Cylinders, facet2=TranType, trans=0.3, n_row=2, n_axis_x_skip=1)`



Nonlinear Fit Line

- `Plot(Weight, MPGhiway, fit="exp", plot_errors=TRUE)`



Regressed linearized data of transformed data values of MPGhiway with `log()`
Line: $b_0 = 4.101$ $b_1 = -0.000$ Linear Model MSE = 0.0095 Rsq = 0.686

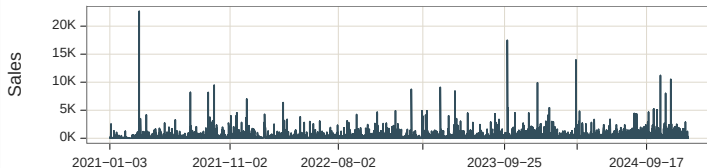
Fit to the data with back transform `exp()` of linear regression model
Model MSE = 9.397178

Tableau Superstore Data

	Variable Name	Type	Values	Missing Values	Unique Values	
1	Order.Date	character	10194	0	1242	... sample values
2	Sales	double	10194	0	5837	
3	Quantity	integer	10194	0	14	
4	Profit	double	10194	0	7334	

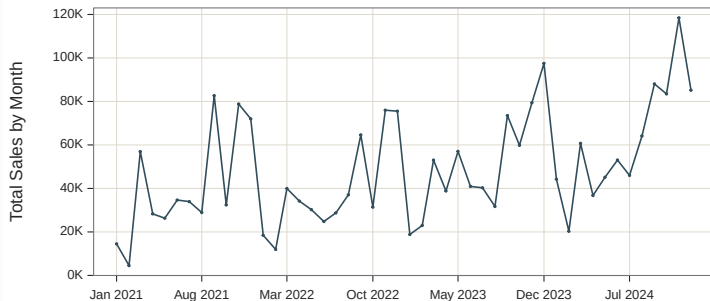
	Order.Date	Sales	Quantity	Profit
1	1/3/21	16.448	2	5.5512
2	1/4/21	3.540	2	-5.4870
3	1/4/21	11.784	3	4.2717
4	1/4/21	272.736	3	-64.7748
5	1/5/21	19.536	3	4.8840

► Plot(Order.Date, Sales)



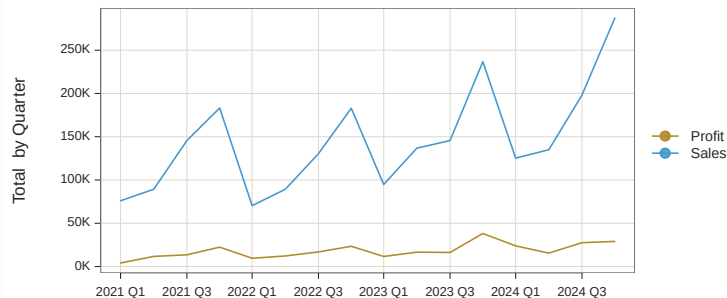
Aggregate over Time

- ▶ Tap into the time series analyses of packages `zoo` and `xts`
- ▶ Aggregate over "days", "months", "quarters", "years"
- ▶ `Plot(Order.Date, Sales, ts_unit="months")`



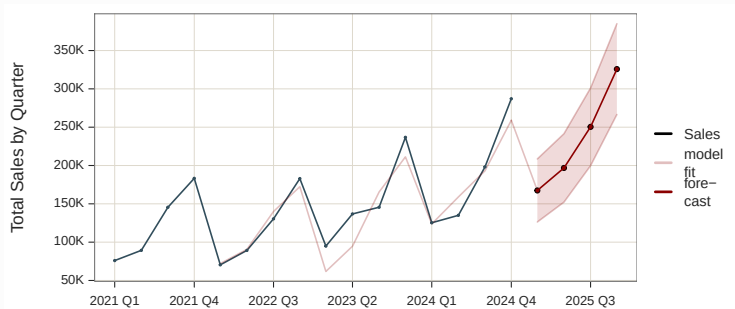
Aggregate Two Variables over Time

► `Plot(Order.Date, c(Sales, Profit), ts_unit="quarters")`



Forecast Future Time Values

► `Plot(Order.Date, Sales, ts_unit="quarters", ts_ahead=4)`



Smoothing Parameters

alpha: 0.2586 beta: 0.6276 gamma: 0.3655

	Order.Date	predicted	upper	lower	width
1	2025 Q1	167420.1	126274.7	208565.6	82290.89
2	2025 Q2	196812.2	152170.9	241453.4	89282.55
3	2025 Q3	250409.9	199728.5	301091.3	101362.76
4	2025 Q4	325794.3	266554.1	385034.4	118480.38

Independent Groups t-test

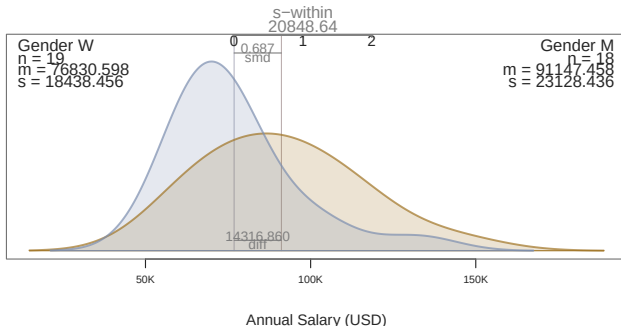
► `ttest(Salary ~ Gender)`

Two-Group Plot with Means

Compare Salary for Gender M and W

Classic t-test of 0 Mean Diff: $t = 2.088$, $df = 35$, $p\text{-value} = 0.044$

95% Confidence Interval for Mean Difference: 395.406 to 28238.314



Compare Salary across Gender with levels M and W

Grouping Variable: Gender, Man or Woman

Response Variable: Salary, Annual Salary (USD)

+ much more

David Gerbing

ANOVA

- ▶ Analysis of Variance with much statistical output and graphics
-
- ▶ One-way: `ANOVA(Y ~ X)`
 - ▶ Independent groups: `ANOVA(Y ~ X + Blocks)`
 - ▶ 2-way Factorial: `ANOVA(Y ~ X1 * X2)`

Linear Regression: Estimated Model

► `reg(Salary ~ Years + Pre + Post, Rmd="eg")`

BASIC ANALYSIS

-- Estimated Model for Salary

	Estimate	Std Err	t-value	p-value	Lower 95%	Upper 95%
(Intercept)	52269.406	14759.452	3.541	0.001	22205.385	82333.426
Years	3289.118	366.884	8.965	0.000	2541.798	4036.437
Pre	-164.842	434.322	-0.380	0.707	-1049.527	719.844
Post	161.756	441.110	0.367	0.716	-736.754	1060.267

-- Model Fit

Standard deviation of Salary: 21,822.372

Standard deviation of residuals: 11,910.713 for df=32

95% range of residuals: 48,522.659 = 2 * (2.037 * 11,910.713)

R-squared: 0.728 Adjusted R-squared: 0.702 PRESS R-squared: 0.640

Null hypothesis of all 0 population slope coefficients:

F-statistic: 28.496 df: 3 and 32 p-value: 0.000

-- Analysis of Variance

Linear Regression: Relations

RELATIONS AMONG THE VARIABLES

-- Correlation Matrix

	Salary	Years	Pre	Post
Salary	1.00	0.85	0.03	-0.05
Years	0.85	1.00	0.05	-0.07
Pre	0.03	0.05	1.00	0.91
Post	-0.05	-0.07	0.91	1.00

-- Collinearity

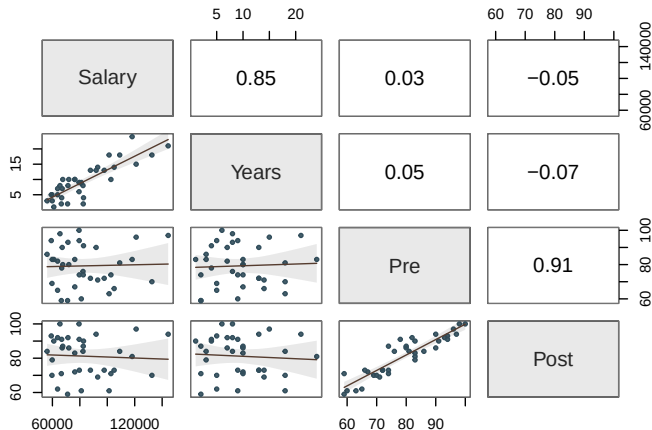
	Tolerance	VIF
Years	0.919	1.088
Pre	0.153	6.551
Post	0.152	6.567

-- Best Subset Regression Models

Years	Pre	Post	R2adj	X's
1	0	0	0.718	1
1	1	0	0.710	2
1	0	1	0.710	2
1	1	1	0.702	3
0	1	1	-0.014	2
0	0	1	-0.026	1

Regression Scatterplot Matrix

► `reg(Salary ~ Years + Pre + Post, Rmd="eg")`



► also a density plot of the residuals and a plot of residuals versus fitted values

Regression: More

- ▶ K-FOLD CROSS-VALIDATION
- ▶ RESIDUALS AND INFLUENCE
- ▶ PREDICTION ERROR
 - ▶ Default Option: For all input data values
 - ▶ Specify input predictor values:
`reg(Y ~ X1 + X2, X1_new=c(5, 10), x2_new=1:5)`
Which provides a grid of all combinations of the specified predictor values
- ▶ Binary logistic regression with a classification matrix
`Logistic(Y ~ X1 + X2)`
where Y is a binary, categorical variable